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Study on the Aggregation Structure and DC Conductivity and Breakdown Characteristics of Polypropylene Insulation for HVDC Cables

Kexin Chen¹  | Tianyi Gao¹ | Man Xu¹ | Yupeng Zhan²  | Shuai Hou² | Mingli Fu² | Yanqun Liao³ | Minhong Lin³

¹State Key Laboratory of Electrical Insulation and Power Equipment, Xi'an Jiaotong University, Xi'an, China | ²Electric Power Research Institute, China Southern Power Grid, Guangzhou, China | ³Zhuhai Power Supply Bureau of Guangdong Power Grid Co. Ltd., Zhuhai, China

Correspondence: Man Xu (xumman@mail.xjtu.edu.cn)

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ABSTRACT

This paper investigates elastomer-toughened polypropylene (PP) insulation to meet the application requirements for green noncrosslinked PP cables in high-voltage direct current (HVDC) transmission. It focuses on the formation of β -crystals in isotactic polypropylene (iPP) by adding a β -nucleator. It examines how varying concentrations of β -nucleator and elastomer (POE) impact the aggregation structure of PP insulation and its conductivity and breakdown characteristics in the DC field. The results indicated that at a β -nucleator agent content of 0.1 wt%, the samples with various POE contents achieved the highest crystallinity, the maximum proportion of β -crystals and the most uniform elastomer distribution. The nucleating agent facilitates the formation of β -crystals in PP and enhances the order degree of the elastomer molecular chains, thereby improving their crystallization capabilities. Evaluations of DC performances and trap characteristics reveal that when the amount of the β -nucleator is set at 0.1 wt%, the sample demonstrates the lowest trap density, an exceptional and lower electric field coefficient of conductivity at elevated electric fields and a superior DC field breakdown strength at 90°C. Compared to samples without β -nucleator, the reduction of DC field breakdown strength for PPBx-0.1 from 25°C to 90°C is approximately 4.86% lower. This improvement is attributed to the ability of the β -nucleator to improve the aggregation structure between PP and POE while optimising the stability of the two-phase interface. Thus, although DC electrical characteristics are maintained at normal temperatures, the DC characteristics are significantly improved at elevated temperatures.

1 | Introduction

Driven by the goal of achieving 'dual-carbon' targets, there is a growing trend towards developing a new power system that prioritises new energy sources. Urbanisation and the construction of large-scale city clusters, clean energy grid access, marine economic development, island power supply and asynchronous grid interconnection all rely heavily on the technical support that high-voltage direct current (HVDC) cables provide. Compared to traditional cross-linked polyethylene (XLPE) insulation cables, green noncross-linked polypropylene (PP)

cables offer several advantages, including lower energy consumption during production, high operating temperature, low space charge accumulation under DC voltage and recyclability [1, 2]. Research studies indicate that the global warming potential (GWP) of PP insulated cables can be reduced by approximately 40% over their entire life cycle [3]. Furthermore, PP does not produce cross-linking by-products that tend to facilitate space charge formation, aligning with the demands for environmentally friendly and low-carbon development in power grids and associated equipment. Furthermore, the long-term operating temperature of XLPE cables is relatively low (AC

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90°C and DC 70°C) [4]. The working temperature of PP insulation cables can be elevated by 10°C–20°C compared with XLPE insulation cables, which can enhance transmission capacity by 10%–20%. However, due to defects, such as high modulus, poor mechanical toughness and low-temperature brittle breaking, the single PP component is not conducive to the production, transportation, laying and use of cables. Its DC electrical performance must also be improved to adapt to the working environment at high temperatures and high voltage [5].

To this end, many scholars have carried out extensive research to regulate the properties of PP through modification, which mainly includes copolymerisation modification, blended modification, grafted modification, nano-doping modification and crystalline modulation. From the analysis of varying technical modifications, each modification method can optimise the performance of PP to a certain extent, but there are also their own shortcomings. Copolymer modification can enhance the mechanical and dielectric properties of PP. However, current copolymer-modified PP requires precise control of the monomer ratio and the polymerisation process during the polymerisation stage to obtain the desired chemical structure while also needing to address the adverse effects of residual catalysts on dielectric properties [6, 7]. Graft modification mainly achieves trap characteristics and distribution regulation by grafting side chains or functional groups onto the polyolefin molecular backbone, thereby improving electrical resistance performance [8–10]. Current research studies primarily focus on grafting maleic anhydride groups onto PP molecular chains, with the main challenges lying in the selection of grafting groups and the control of the grafting rate/process. Nanocomposite modification improves space charge characteristics and breakdown performance by incorporating different nanoparticles to introduce interface regions that regulate charge transport processes [11]. However, nanoparticles restrict molecular chain movement, leading to increased material hardness [12], and face challenges in achieving uniform dispersion, which limits their practical application in cable insulation.

Among these methods, incorporating elastomers with superior molecular chain flexibility is particularly effective in reducing the modulus of elasticity and increasing the elongation at the break of PP. In addition, elastomer blending is one of the most straightforward approaches for industrial-scale production, making PP achieve mechanical properties (such as hardness and brittleness at low temperatures) closest to those of XLPE [13]. However, because the elastomer itself is often a weak area of electrical breakdown compared to PP, it is easy to have defects in the weak phase interface formed with PP, which may cause a decrease in electrical strength in the modified material system [14, 15]. Therefore, scholars are devoted to enhancing the insulation performances of PP/elastomer blends [16]. Furthermore, as the working temperature increases, the molecular chains of the elastomer first experience intense thermal movement, which may disrupt the phase-interface structure, lead to aggregation and consequently impact the high-temperature mechanical and electrical properties of the material.

Numerous studies have shown that PP with β -crystals has excellent mechanical properties, and β -crystals have a positive effect on the enhancement of the electrical properties of PP

under the premise of effectively improving the mechanical properties of PP [17, 18]. Previous research studies by the authors revealed that introducing β -crystals into PP improved the stability of electrical properties in response to temperature variations, as both the electrical field breakdown strengths at room temperature and 90°C were significantly improved [19–21]. Meanwhile, some scholars have investigated the synergistic effects of β -nucleating agents and elastomers on the performance of PP modification systems. It has been found that this modification approach exerts positive effects on both the mechanical and electrical properties of composites. However, existing studies predominantly focus on enhancing mechanical properties [22, 23] or investigating electrical performance under AC electric fields [24] and at room temperature [25]. Based on these results, this study explores elastomer-toughened PP insulation, which possesses excellent mechanical properties. The aggregation structure of the PP/elastomer system was further optimised through crystalline modulation by adding a β -nucleator agent. This study investigates the impact of β -crystals on isotactic polypropylene (iPP) and polyolefin elastomer (POE)-blended systems under conditions of high temperature and high electrical field.

2 | Experimental Methods

2.1 | Sample Preparation

The iPP manufactured by Maoming Petrochemical and branded PPH-T03 has a melt index of 2.5 g/10 min and an isotacticity of approximately 96%. The POE produced by Dow (USA) is identified with the brand name POE8450 and has a melt index of 3.0 g/10 min. The manufacturer of the rare earth β -nucleator agent WBG-II is Guangdong Weilin Co.

The raw materials were dried in a vacuum oven at 70°C for 12 h to prepare them. Following this, a melt blending and pelletising process was conducted using German micro-twin-screw extruder equipment (Hakke Process11). The temperature settings for the extruder were calibrated from the inlet to the extrusion die port as follows: 190°C, 195°C, 200°C, 200°C, 200°C, 200°C, 200°C, 200°C and 200°C in that order. To achieve optimal dispersion of β -nucleator agent within the PP matrix, a master batch was first prepared by melt blending iPP and WBG in a mass ratio of 95:5. Subsequently, iPP/WBG samples with varying WBG contents were prepared by melt blending the master batch and iPP pellets. The iPP/WBG and POE pellets were then melt blended to form iPP/WBG/POE samples with different POE contents. Additionally, iPP and POE with various contents were extruded and pelletised using the twin-screw extruder under the same conditions, with the iPP/POE material system serving as a comparative sample. The names of each sample are detailed in Table 1; PPB means PP/POE blends. Finally, the samples were pressed into

TABLE 1 | Name of each sample.

Sample name	POE content (wt%)	WBG content (wt%)
PPBx-y	x	y

Note: x = 20, 30, or 40, and y = 0.05, 0.1, 0.3, or 0.5.

sheets with thicknesses of 0.2 and 1 mm using a flatbed vulcanising machine at 200°C and 10 MPa.

2.2 | Structural Characterisation

The crystal characteristics of the samples were examined using X-ray diffraction (XRD) with a BRUKER D8 ADVANCE25 model instrument, utilising a CuK α target (wavelength $\lambda = 0.154$ nm). The experimental parameters included a voltage of 40 kV, a current of 40 mA, a scanning range of 10°–30°, a step size of 0.2° and testing temperatures of 25°C and 90°C.

Differential scanning calorimetry (DSC) was employed to investigate the melting and crystallization behaviour of the samples under a nitrogen atmosphere. Each sample weighed approximately 6 mg. The temperature was initially raised to 220°C at 10°C/min and held for 5 min to eliminate thermal history. Then, the temperature was decreased to 50°C at a rate of 10°C/min to obtain the crystallization curve, followed by a subsequent increase to 220°C at 10°C/min to obtain the melting curve.

Dielectric characterisation was performed on a Concept43 spectrometer, a system designed for measurements across wide frequency ranges under elevated temperatures and high electric fields. The experimental settings of the frequency range were 10^{−2}–10⁴ Hz, and the temperatures were 25°C and 90°C, with an applied voltage of 1 kV. The sample thickness was approximately 1 mm. To ensure optimal contact between the sample and the electrodes, the surface of the sample in contact with the upper electrode was spray-plated with a circular gold electrode of 30 mm in diameter. In comparison, the surface of the sample in contact with the lower electrode was spray-plated with a circular gold electrode of 40 mm in diameter.

2.3 | DC Performance Testing

Thermally stimulated depolarisation current (TSDC) was employed to investigate the trap distribution characteristics of a material with a sample thickness of (0.20 ± 0.02) mm. Initially, the sample was rapidly heated to a polarization temperature of 80°C from room temperature, during which an electric field of 1 kV/mm was applied to maintain the polarization for 20 min. Following the completion of polarization, the temperature of the sample was lowered to −75°C at a cooling rate of −30°C/min, designated as the depolarisation temperature, where it was stabilised for 5 min. Finally, the sample was elevated to the specified temperature value of 125°C at a heating rate of 2°C/min, and the current during this heating process was recorded to calculate the trap parameters of the sample.

The conductivity characteristics were assessed using a Model 6517A high resistance metre, a SPELLMAN SL300 DC power supply and a custom-built standard three-electrode system with shielding. The electric field strengths applied were 30, 40 and 50 kV/mm, with temperatures set at 25°C and 90°C while maintaining the sample thickness of (0.20 ± 0.02) mm. The electrode system and sample were placed in an oven during

testing. After reaching the desired temperature, a waiting period of 120 min was observed before initiating pressurisation. Current measurements were recorded during a 20-min pressurisation period, and the median of the current signals over the last 60 s was used to calculate the conductivity parameters.

The voltage breakdown tester (GJW-100E) was employed to assess the DC breakdown field strength of the samples, using a ball–ball electrode with a diameter of 25 mm. The experiments were conducted at 25°C and 90°C, with the sample thickness maintained at (0.20 ± 0.02) mm. Both the electrode and the sample were immersed in a temperature-controlled chamber filled with silicone oil, with a temperature variation not exceeding 3°C during testing. The voltage was increased to 1 kV/s until the samples were broken down. Each sample was broken down no less than 15 times, and the resulting electrical breakdown field strength values were analysed using Weibull distribution probabilistic statistical theory.

3 | Results and Discussion

3.1 | Crystal Characterisation

To investigate the effect of introducing elastomer and β -nucleator agent on the crystalline structure of PP, the XRD spectra were recorded for each sample at 25°C and 90°C, as illustrated in Figures 1 and 2, respectively. The analysis revealed that diffraction peaks appeared in all samples at diffraction angles 2θ of about 14.05°, 16.90°, 18.60°, 21.10° and 21.86°. These peaks correspond to the α (110), α (040), α (130), α (131) and α (111) crystal planes of α -crystals in PP, respectively. Notably, with the introduction of the β -nucleator, a diffraction peak for the β -crystals was identified at a 2θ angle of 16.05°, corresponding to the β (300) crystal plane. The proportion of β -crystals to total crystalline content in PP, K_β , was calculated using DIFFRAC. eva V3.1 software according to Equation (1) proposed by Turner-Jones et al.

$$K_\beta = \frac{H_{300}}{H_{300} + H_{110} + H_{040} + H_{130}} \times 100\%, \quad (1)$$

where H_{300} , H_{110} , H_{040} and H_{130} are the diffraction peak intensities of crystal faces (300), (110), (040) and (130), respectively. The crystallinity of each sample can also be obtained, as shown in Table 2.

Analysing the diffractogram of each sample leads to the following conclusions. (1) As the contents of elastomer increase, the intensity of characteristic diffraction peaks of β -crystals in the sample shows a decreasing trend. This suggests that the elastomer has a certain inhibitory effect on the nucleation efficiency of the β -nucleator agent. (2) At the same POE content, increasing the contents of β -nucleator results in the content of β -crystals in the samples to initially increase and then decrease. Specifically, when the β -nucleator content is at 0.05 wt%, an increase in elastomer content causes a decrease in K_β of the samples. However, with higher β -nucleator content, the changes in K_β are minimal, indicating that the inhibitory effect of elastomers on the nucleation efficiency is weakened. (3) Following the addition of the β -nucleator, the pattern of crystallinity

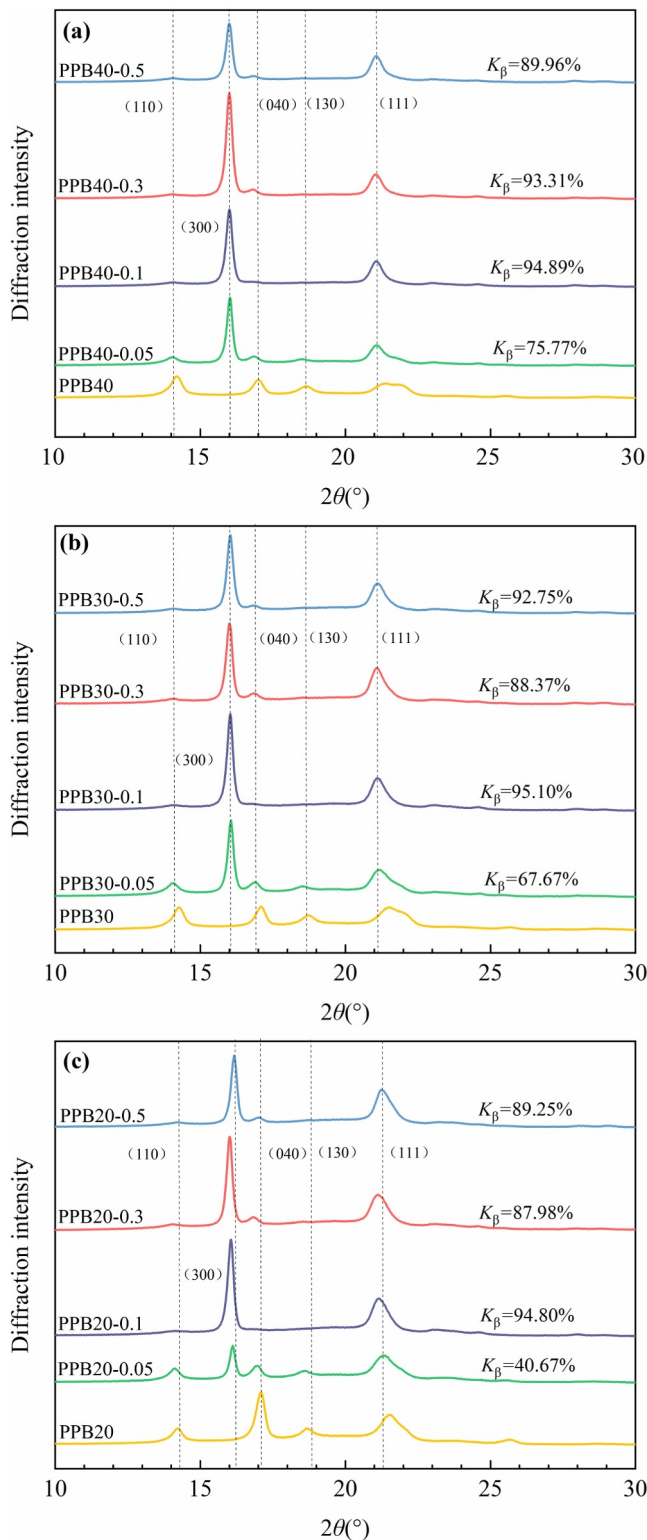


FIGURE 1 | XRD diffractogram of each sample at 25°C: (a) PPB20-y, (b) PPB30-y, (c) PPB40-y.

changes in each sample closely mirrors the variations in β -crystals' content. Notably, the crystallinity of the PPBx-0.05 system is the lowest, suggesting that a lower β -nucleator can lead to reduced crystalline conversion efficiency, reducing the overall crystallinity and negatively impacting the insulation performance of the system. (4) The diffractogram of the sample

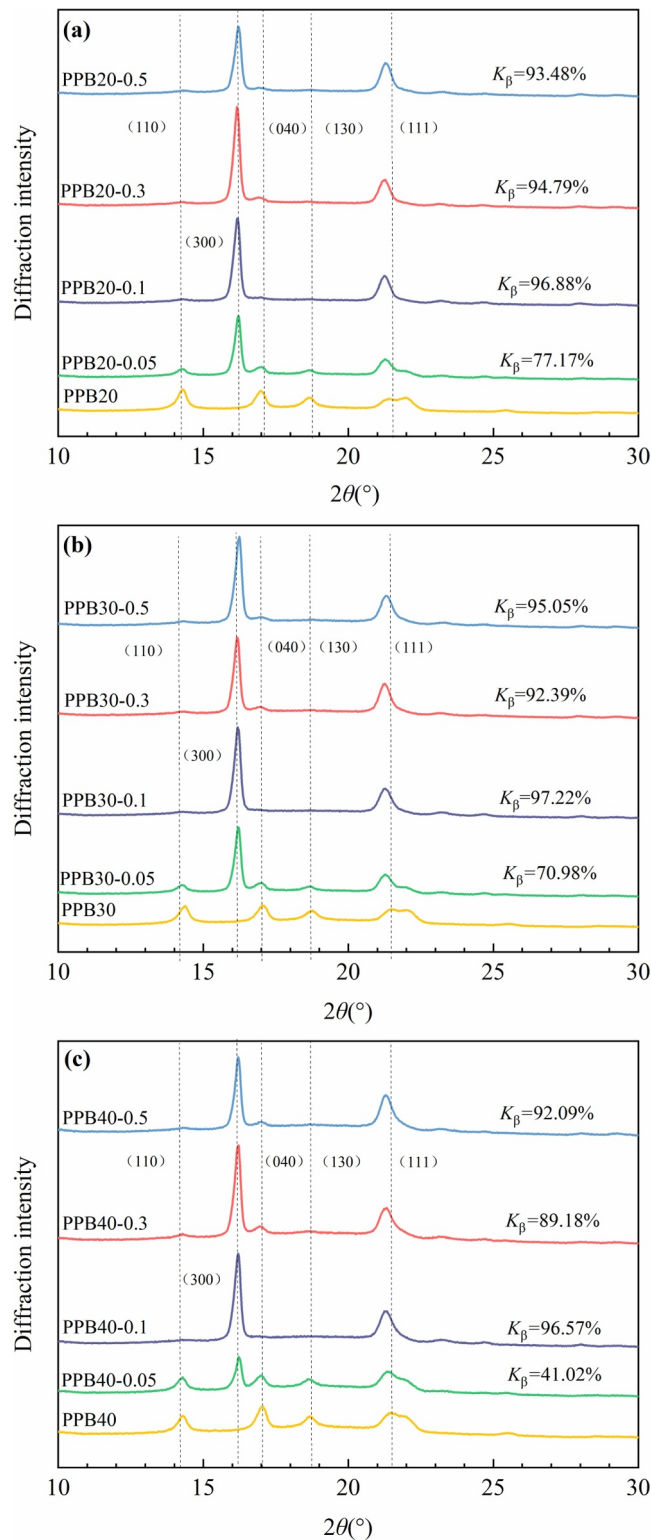


FIGURE 2 | XRD diffractogram of each sample at 90°C: (a) PPB20-y, (b) PPB30-y, (c) PPB40-y.

in the morphology of the characteristic peaks and the law of change at 90°C are the same as at 25°C. However, the crystallinity of the samples has been reduced at 90°C without the β -nucleator. Upon addition of the β -nucleator, the content of β -crystals and the degree of crystallinity increase. This suggests that a small portion of α -crystalline continues to transform into

TABLE 2 | Crystallinity of each sample obtained by XRD.

Samples	X_C (%)	
	25°C	90°C
PPB20	39.7	38.3
PPB20-0.05	38.2	37.3
PPB20-0.1	42.7	43.9
PPB20-0.3	43.2	43.9
PPB20-0.5	40.0	40.7
PPB30	38.5	34.7
PPB30-0.05	35.8	31.6
PPB30-0.1	39.5	40.0
PPB30-0.3	38.2	39.3
PPB30-0.5	38.4	38.0
PPB40	38.0	31.7
PPB40-0.05	32.1	27.9
PPB40-0.1	38.7	35.5
PPB40-0.3	34.8	33.9
PPB40-0.5	36.8	33.8

β -crystalline as the temperature rises within the system. Moreover, the elastomer phase is enhanced at elevated temperatures (inhibition of thermal movement of the molecular chains of elastomer), increasing the overall crystallinity.

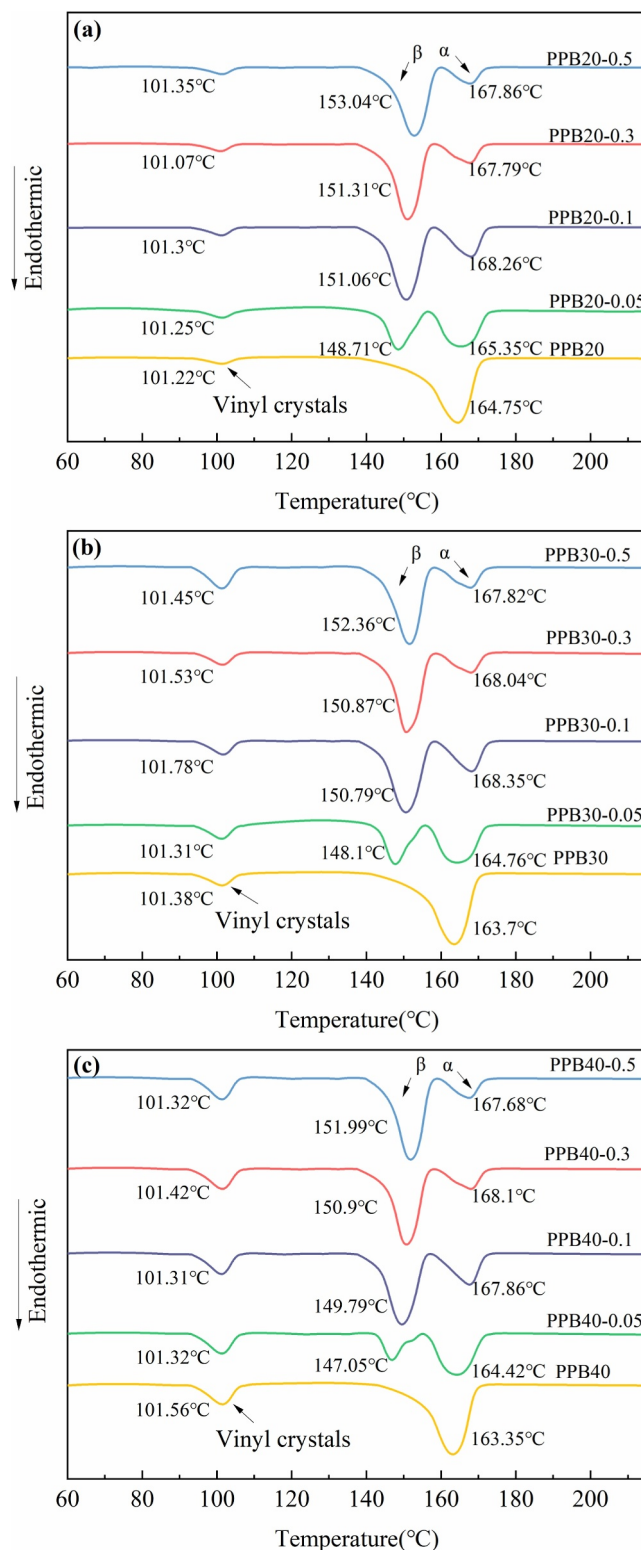
The elastomer phase is the weak point within the iPP/POE-toughened system. The introduction of β -nucleator positively influences the ordering of elastomer chains in the composite phase, reinforcing this weak point. Notably, when the content of β -nucleator is at 0.1 wt%, the sample exhibits the highest content of β -crystals, which remains stable even as the elastomer content increases. Simultaneously, the crystallinity of the PPB-x-0.1 system is also the largest.

3.2 | Crystal Characterisation

The melting and crystallization curves of various samples are presented in Figures 3 and 4, respectively. In the PPBx samples without the addition of β -nucleator there were two melting peaks: one was the melting peak of α -crystals near 165°C, and the other was the melting peak of ethylene crystals of POE near 101°C. However, with the introduction of β -nucleator, a third peak emerged in PPBx-y samples, indicating the melting of β -crystals around 150°C. The crystallinity X_C was calculated as detailed in Equation (2), wherein the crystallinity of each sample is the cumulative total of its ethylene crystals, α -crystals and β -crystals crystallinity. The results of the calculations are shown in Table 3.

$$X_C = \frac{\Delta H_m}{\Delta H_m^0}, \quad (2)$$

where ΔH_m is the actual enthalpy of melting of the sample; ΔH_m^0 is the standard melting enthalpy of the sample at 100% crystallinity. The standard enthalpy of melting for polypropylene

**FIGURE 3** | Melting curve of each sample: (a) PPB20-y, (b) PPB30-y, (c) PPB40-y.

α -crystals is 177.0 J/g, for β -crystals is 168.5 J/g and for ethylene crystals is 288.4 J/g.

The analysis of the melting curves reveals that as the content of POE increases, the melting peak area of ethylene crystals within POE also increases accordingly. At the same time, the overall

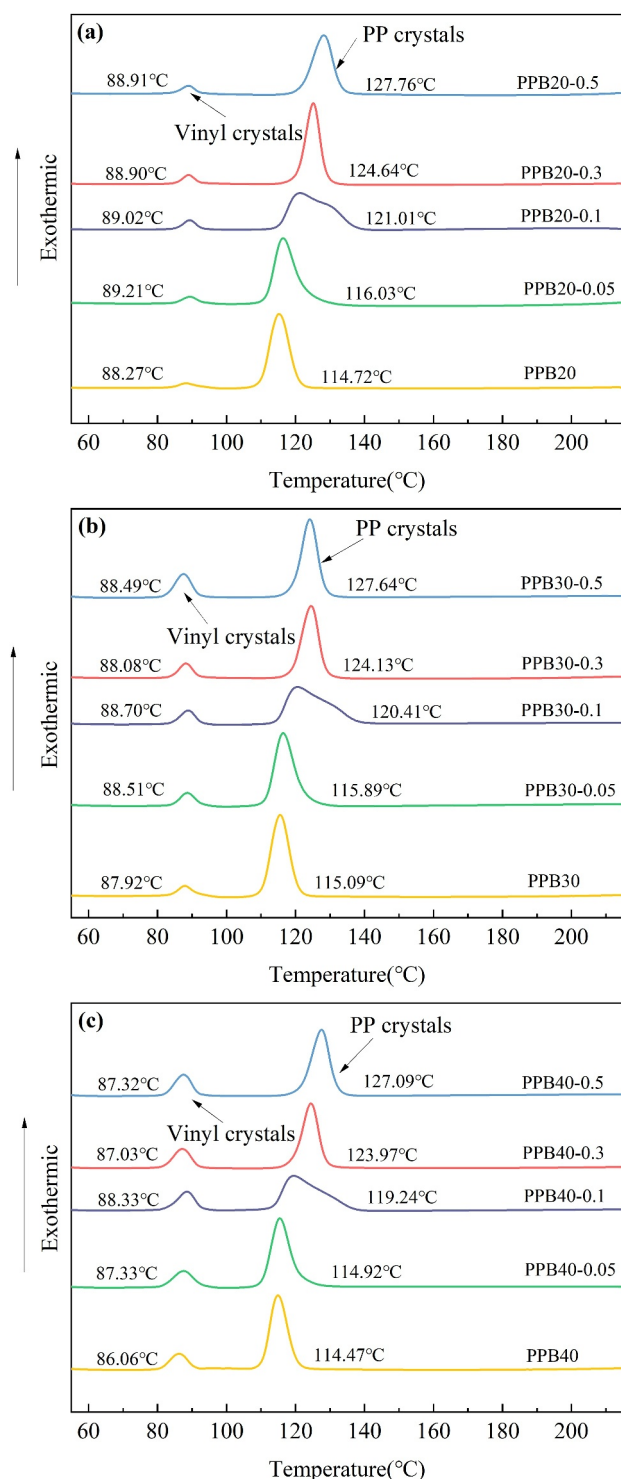


FIGURE 4 | Crystallization curve of each sample: (a) PPB20-y, (b) PPB30-y, (c) PPB40-y.

crystallinity of each sample gradually declines. At the same POE content, an increase in the nucleating agent results in the following observations. (1) The melting peak area of ethylene crystals in POE increases, indicating that the nucleating agent plays a significant role in promoting the crystallization of elastomers. (2) The melting temperature of α -crystals initially increases and then decreases, with the melting peak area of α -crystals gradually decreasing, indicating a more frequent conversion of α -crystals to β -crystals. (3) The melting temperature

TABLE 3 | Crystallinity of each sample obtained by DSC.

Samples	X _C (%)
PPB20	32.59
PPB20-0.05	26.66
PPB20-0.1	40.60
PPB20-0.3	36.81
PPB20-0.5	32.39
PPB30	31.59
PPB30-0.05	24.32
PPB30-0.1	34.28
PPB30-0.3	30.69
PPB30-0.5	31.49
PPB40	23.03
PPB40-0.05	19.16
PPB40-0.1	32.00
PPB40-0.3	27.83
PPB40-0.5	29.98

of β -crystals displays an upward trend with the melting peak area of β -crystals first increasing and then decreasing. Notably, the peak area of β -crystals is the largest when the content of β -nucleator is 0.1 wt%. Consequently, with the addition of a nucleating agent, β -crystals become the predominant form. As the β -nucleator content increases, the crystallinity also exhibits a trend of initially increasing and then decreasing, with the crystallinity of PPBx-0.1 being the highest.

The crystallization curves reveal that the crystallization temperatures of ethylene and PP crystals slightly decreased as the POE content increased with only minor changes observed. For a given POE content, the crystallization temperature of PP crystals gradually increased, with higher levels of β -nucleator. Notably, at a nucleator concentration of 0.1 wt%, the width of the crystallization peaks reached the maximum. The crystallization behaviour of the composite system reveals the following insights. (1) In the absence of β -nucleator, only α -crystals were present in the PP crystals, represented by the PPBx curves, and the crystallization temperature was relatively low. (2) With the addition of the β -nucleator, both α and β crystals emerged in the PP crystallization process. The presence of β -crystals enhanced the crystallization temperature of PP; however, at a low nucleating agent content of 0.05 wt%, the limited quantity of β -crystals had minimal impact on the overall PP crystallization pattern. (3) Increasing the nucleating agent content to 0.1 wt% resulted in XRD observations that indicated β -crystals constituted more than 94% of the crystallization. β -crystals dominate the PP crystallization process at this concentration. The longer crystallization peak width at this context suggests that β -crystals form first during the crystallization process of PP, followed by the formation of α -crystals at a lower temperature. (4) When the content of the nucleating agent continues to increase, the characteristic of a wide crystallization peak disappears. This indicates that the crystal transformation efficiency in PP has reached a critical value, where the crystallization of β -crystals becomes dominant, with only a tiny number of α -crystals

remaining untransformed. Consequently, the crystallization temperature further increases. This also suggests that the role of the β -nucleator is more aligned with facilitating crystallographic transformation rather than generating new crystals.

Overall, as the content of the β -nucleator increases, the impact of enhanced POE content on the crystallinity is significantly reduced compared to the systems without the nucleating agent. The inhibitory effect of the elastomer on the nucleation efficiency of the nucleating agent previously observed becomes less pronounced. The β -nucleator can enhance the crystallization capacity of the elastomer, with the maximum crystallinity of the system occurring at a β -nucleator content of 0.1 wt%. This is consistent with the results of the XRD analysis.

3.3 | Dielectric Characterisation

PP is a nonpolar material where electron displacement polarization predominates when subjected to an alternating electric field. The formation time for electron displacement polarization is extremely short, typically ranging from 10^{-15} to 10^{-13} s (below 10^{15} Hz), which renders the dielectric constant of PP effectively independent of frequency. Modified PP insulation also exhibits space charge and interface polarization with longer establishment times. These types of polarization primarily influence the dielectric properties within the frequency range of 10^{-5} – 10^6 Hz. The present experiment mainly examines the polarization characteristics of the low-frequency band to reflect the interfacial structural characteristics of PP insulation. Table 4 presents the variation of dielectric constant with temperature for each sample, comparing those without β -nucleator to those with 0.1 wt% β -nucleator.

At 25°C, an increase in the content of POE corresponds to an increase in the dielectric constant of each sample. This phenomenon is mainly attributed to the interface between the PP and POE phases. The significant conductivity difference between the two phases leads to charge accumulation at their interface under the influence of an external electrical field, which triggers interfacial polarization. The heightened interfacial polarization results in an increased dielectric constant. The addition of the β -nucleator does not significantly alter the dielectric constant, suggesting that while it modifies the crystalline structure of PP and influences the morphology of the

two-phase interface, it does not significantly impact the polarization and dielectric constant of the system at room temperature. Furthermore, as the frequency increases from 10^{-2} – 10^4 Hz, the dielectric constant of each sample decreases. This indicates that the movement of dipole groups in the electric field is insufficient to keep pace with the frequency changes, reflecting a reduction in the polarization at the interface between the two phases.

At 90°C, the dielectric constant of each sample at 10^4 Hz decreases significantly. This reduction is primarily attributed to the volume expansion of the material at elevated temperatures, which leads to a decrease in the number of molecules per unit volume and a corresponding reduction in the degree of polarization. Conversely, the dielectric constant of each sample at 10^{-2} Hz shows a significant increase. This is primarily attributed to enhanced interfacial polarization and accelerated segmental mobility of polymer chains. That is to say, elevated thermal energy promotes charge migration towards interfaces (spherulite boundaries and elastomer/PP domains), intensifying interfacial polarization. Additionally, increased free volume and chain mobility facilitate dipole alignment under low-frequency fields. Notably, the dielectric constant of PPBx-0.1 is lower than that of PPBx. Analysing the ratio of the dielectric constants of each sample at 90°C and 25°C at 10^{-2} Hz reveals that the system with a β -nucleator exhibits higher temperature stability in its dielectric constants. This suggests that the introduction of a β -nucleator enhances the stability of the two-phase structure of PP and POE at high temperatures.

3.4 | Trap Distribution Characterisation

According to the TSDC theory, the trap-induced thermal stimulation current can be expressed as follows:

$$I(T) = A \exp\left(-\frac{E_t}{kT}\right) \times \exp\left(\frac{B}{\beta_1} \int_{T_0}^T \exp\left(-\frac{E_t}{kT}\right) dT\right), \quad (3)$$

where $I(T)$ is the TSDC measured current, A; T is the temperature, K; A is a constant; E_t is the trap energy level depth measured by TSDC, eV; k is the Boltzmann's constant, 1.381×10^{-23} J; B is a constant; β_1 is the rate of temperature increase, K/min; T_0 is the starting temperature, K.

TABLE 4 | Dielectric constant of each sample.

Samples	Dielectric constant at 10^{-2} Hz		Dielectric constant at 10^4 Hz		Ratio of the dielectric constants of 90°C and 25°C at 10^{-2} Hz
	25°C	90°C	25°C	90°C	
PPB20	2.49	2.60	2.45	2.38	1.044
PPB30	2.50	3.28	2.47	2.39	1.312
PPB40	2.51	3.84	2.46	2.38	1.530
PPB20-0.1	2.51	2.74	2.45	2.36	1.092
PPB30-0.1	2.50	3.11	2.45	2.35	1.244
PPB40-0.1	2.52	3.61	2.47	2.34	1.433

When the temperature is lower, $T \rightarrow T_0$ then:

$$\int_{T_0}^T \exp\left(-\frac{E_t}{kT}\right) dT \rightarrow 0, \quad (4)$$

$$I(T) = A \exp\left(-\frac{E_t}{kT}\right), \quad (5)$$

Taking logarithms around Equation (5), as in Equation (6), and making a linear fit, the trap energy level depth E_t of the composite can be obtained by the slope of the straight line.

$$\ln I(T) = \ln A - \frac{E_t}{kT}. \quad (6)$$

The TSDC curves of each sample are illustrated in Figure 5, accompanied by the calculated trap energy levels (E_t) and the incoming charge densities (N_t) for each sample, as detailed in Table 5. According to the related literature [26–28], it can be observed that the current peaks near 60°C in the TSDC curves of PP are induced by the molecular chains confined in the amorphous region by the crystalline region during the process of depolarisation. In contrast, the molecular chains that are arranged and ordered near the crystalline region lead to traps with higher energy levels.

Figure 5 reveals that for the toughened system without a β -nucleator, the peak temperature of traps in each sample shifts gradually towards lower temperature with an increase in POE content, accompanied by a corresponding decrease in trap energy levels. This phenomenon is primarily due to the higher elastomer content that disrupts the crystalline structure of PP to some extent. In this system, the traps arise mainly from the α -inter-crystalline molecular chains of PP, elastomer molecular chains and the defects at the interface of the two phases. This results in a overall higher trap density.

In the toughened system using a β -nucleator, β -crystals predominantly formed at elevated temperatures during the crystallization, followed by the formation of α -crystals at lower temperatures. At this stage, the energy level and distribution of traps are influenced by the α -crystals and β -crystals within PP and the elastomer. Our previous research study [20] indicates that, compared with α -crystals, the parallel arrangement of the β -crystal wafer structure results in a weaker coupling of PP molecular chains in the interface region between β -crystals (nanoscale), and a reduced binding force of PP molecular chains in the amorphous region. The elevated relative content of β -crystals, along with the formation of ‘flower-like’ β -crystal clusters, leads to a reduction in deep trap energy levels and an increase in trap charge density.

This study shows that introducing a β -nucleator reduces the trap energy level across all samples due to crystalline transformation. However, the overall incoming charge density also reduces. The reasons can be summarised as follows: β -PP in the matrix induced by the crystalline modulation increases the trap density. However, the increase in the overall proportion is minimal. It will also increases the degree of order of the elastomer molecular chain, optimising the structure defects caused by the elastomer to reduce the number of defects and to

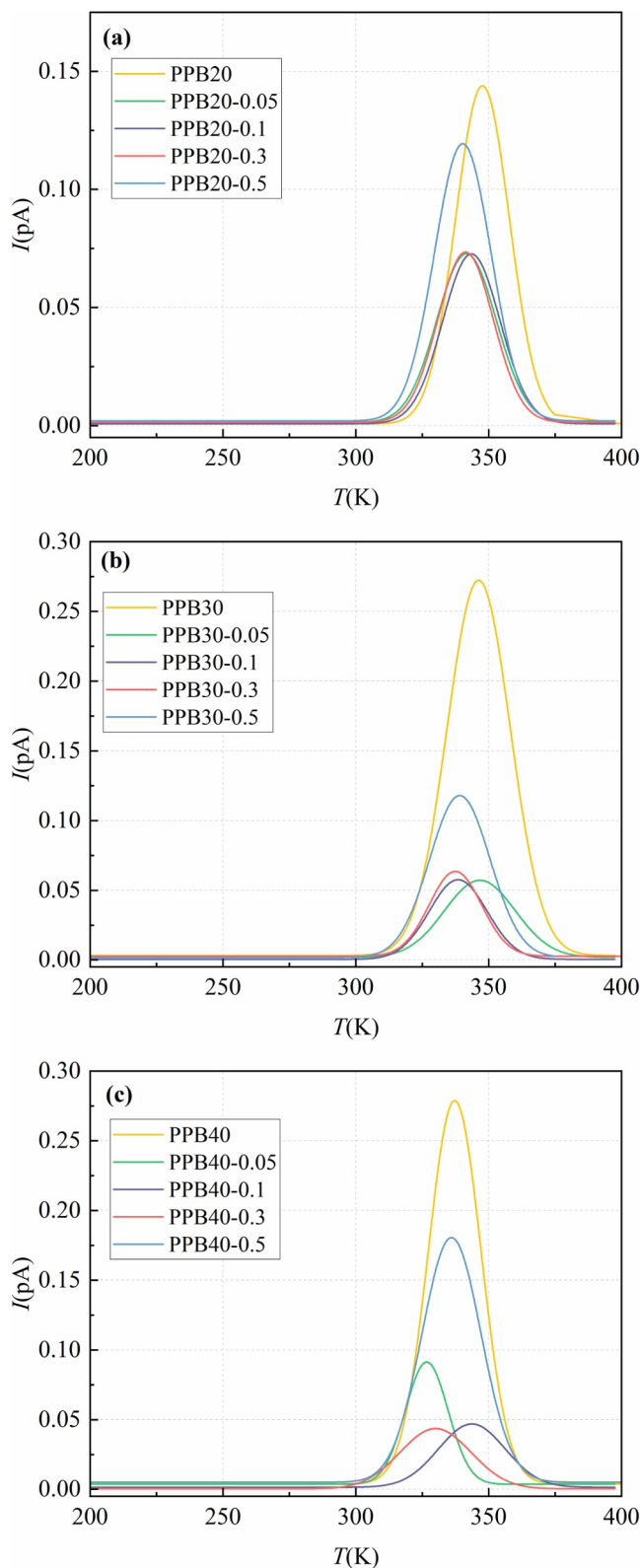


FIGURE 5 | Characteristic curve of depolarisation current for each sample: (a) PPB20-y, (b) PPB30-y, (c) PPB40-y.

decrease the defect density overall. Notably, the lowest trap density of the sample occurs at a β -nucleator content of 0.1 wt%. This finding further highlights the significant impact of the β -nucleator on the results of the XRD, DSC, and low-frequency dielectric response measurements.

TABLE 5 | Trap energy levels and incoming charge density for each sample.

Samples	E_t (eV)	$N_t (\times 10^{16} \text{ m}^{-3} \text{ eV}^{-1})$
PPB20	1.33	5.98
PPB20-0.05	1.21	3.24
PPB20-0.1	1.19	2.92
PPB20-0.3	1.03	3.45
PPB20-0.5	0.95	5.53
PPB30	1.22	13.52
PPB30-0.05	1.23	3.48
PPB30-0.1	1.21	2.65
PPB30-0.3	0.99	3.18
PPB30-0.5	0.93	5.69
PPB40	1.18	11.97
PPB40-0.05	0.83	3.90
PPB40-0.1	0.78	2.15
PPB40-0.3	0.76	2.67
PPB40-0.5	0.80	3.72

3.5 | Conductivity Characteristics

Generally, the DC conductivity of insulating materials is influenced by temperature and the applied electric field. The empirical formula for DC conductivity can be expressed by Equation (7). At a certain temperature, the relationship between DC conductivity and electric field can be expressed by Equation (8).

$$\gamma = \gamma_0 e^{\alpha T} E^{\beta_2}, \quad (7)$$

$$\gamma = CE^{\beta_2}, \quad (8)$$

where γ represents the DC conductivity at a given temperature T and electric field E , γ_0 represents the DC conductivity at the reference temperature and electric field, α represents the temperature coefficient, β_2 represents the electric field coefficient, C is a constant.

Linear fitting of the conductivity data under high electric field strengths (the tested electric field strengths are 30, 40 and 50 kV/mm, respectively) reveals that it conforms to the pattern described by Equation (8). The variation in conductivity with electric field strength for each sample at 25°C and 90°C is depicted in Figure 6. The electric field coefficient β_2 for each sample can also be derived. It is evident that as the POE content increases, the conductivity of the samples increases significantly, with nearly a one-order-of-magnitude increase at room temperature when the POE content reaches 40 wt%. This is due to the decrease in the crystallinity of the system, which causes the distribution of crystalline zones to become looser, promoting charge transport and increasing the conductivity.

Upon the addition of the β -nucleator, the conductivity of the samples decreased, and the electric field coefficient β_2 of each sample was reduced. This reduction is primarily attributed to β -crystals' role in refining spherulite size and increasing

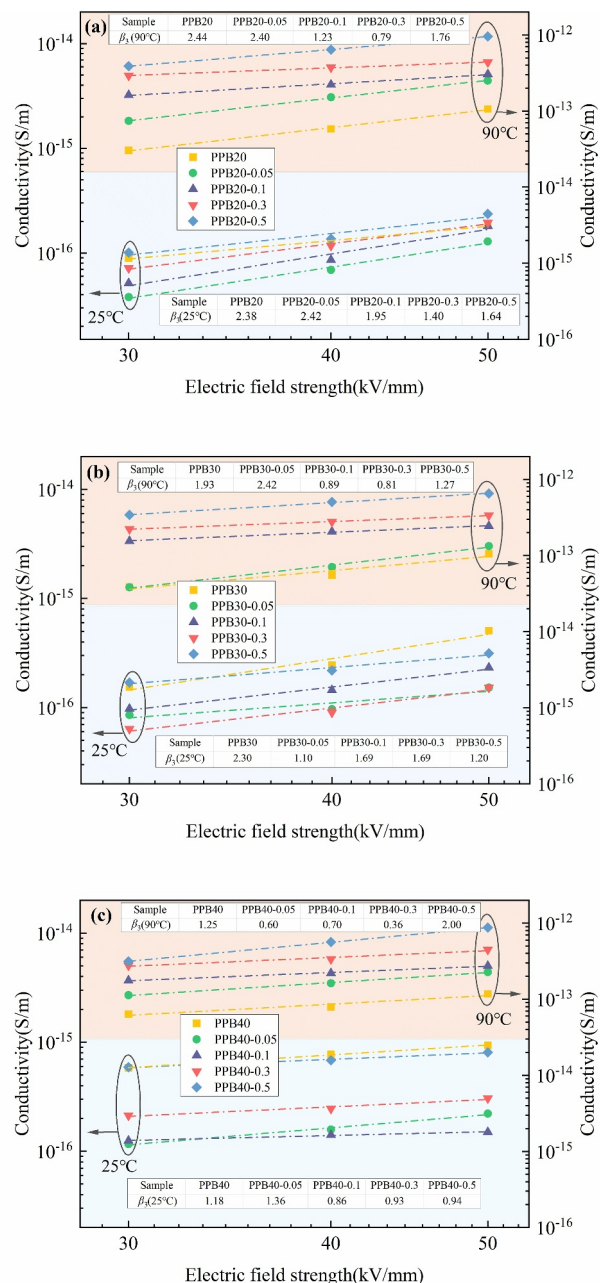


FIGURE 6 | Conductivity-electric field strength characteristic curves of each sample at different temperatures: (a) PPB20-y, (b) PPB30-y, (c) PPB40-y.

interfacial area between PP and elastomer (microscale), which can make more interfaces between the crystals compared with α -crystals, improving compatibility between the two phases and interfacial stability. This configuration enhances the role of elastomer chain segments and the PP crystals at the crystal boundaries, effectively inhibiting electric charge transport. At a constant concentration of POE, an increase in the β -nucleator content leads to an increase in conductivity. When the β -nucleator content is at 0.5 wt%, it appears to have no inhibitory effect on the charge transport in the elastomer-toughened system. However, at a β -nucleator concentration of 0.1 wt%, the conductivity of the samples remains low, and only slight changes in conductivity are observed with an increase in electrical field strength.

At elevated temperatures (90°C), the free volume in the amorphous region of PP increases, facilitating easier charge migration. The gradual loosening of the bonds between the elastomer chain segments and the PP phases further promotes charge transport within the samples, ultimately increasing their electrical conductivity. After adding the β -nucleator, the conductivity of each sample at elevated temperatures decreased. This is because the coupling between β -crystals is relatively weaker compared to that of α -crystals, and the number of amorphous chain molecules in β -crystals is greater than that in α -crystals, which can be explained by the reduction of deep trap energy levels in the TSDC analysis. However, when the content of the β -nucleator was at 0.1 wt% or 0.3 wt%, the high-temperature and high-field-strength conductivity of the samples were still lower than those of the single iPP/POE-toughened system. In other words, the conductivity characteristics of the samples exhibited greater stability with variations in electric field strength.

Based on previous results in this study, it can be concluded that the PPBx-0.1 sample contains the highest proportion of β -crystals and exhibits the highest crystallinity. At this juncture, the β -nucleator significantly enhances the stability of the POE phase itself and the interface between the PP and the POE phases at high temperatures. This enhancement of POE segments promotes the formation of denser crystalline regions within the polymer, inhibiting charge transport in the amorphous region, and ultimately improves the DC conductivity performance of the samples, especially their conductivity characteristics at elevated temperatures and under high electric fields.

3.6 | DC Breakdown Characteristics

The DC breakdown test data is analysed using the Weibull distribution, which facilitates determining the probability of breakdown of insulation materials under the electric field strength E as expressed by Equation (9).

$$P(E) = 1 - \exp\left[-\left(\frac{E}{E_0}\right)^{\beta_3}\right], \quad (9)$$

where P is the cumulative breakdown probability; E is the test breakdown strength; E_0 is the scale parameter, the breakdown strength when the breakdown probability is 63.2%, also known as the characteristic breakdown strength; β_3 is the shape parameter, which can be used to reflect the dispersion of the data, and the larger the value, the smaller the dispersion. The Weibull distribution diagrams of the DC breakdown strength of the sample at 25°C and 90°C are shown in Figure 7, and Table 6 shows the parameters of the Weibull distribution of the DC breakdown of the sample.

At 25°C, the DC breakdown strength of the samples shows minimal variation. However, as the POE content increases, the breakdown strength of each sample decreases. This reduction is attributed to the inherent limitations of the electrical strength of the elastomer, the decrease of crystallinity and an increase of the interfacial area, which provides smoother spatial channels for carrier migration and then facilitates the formation and

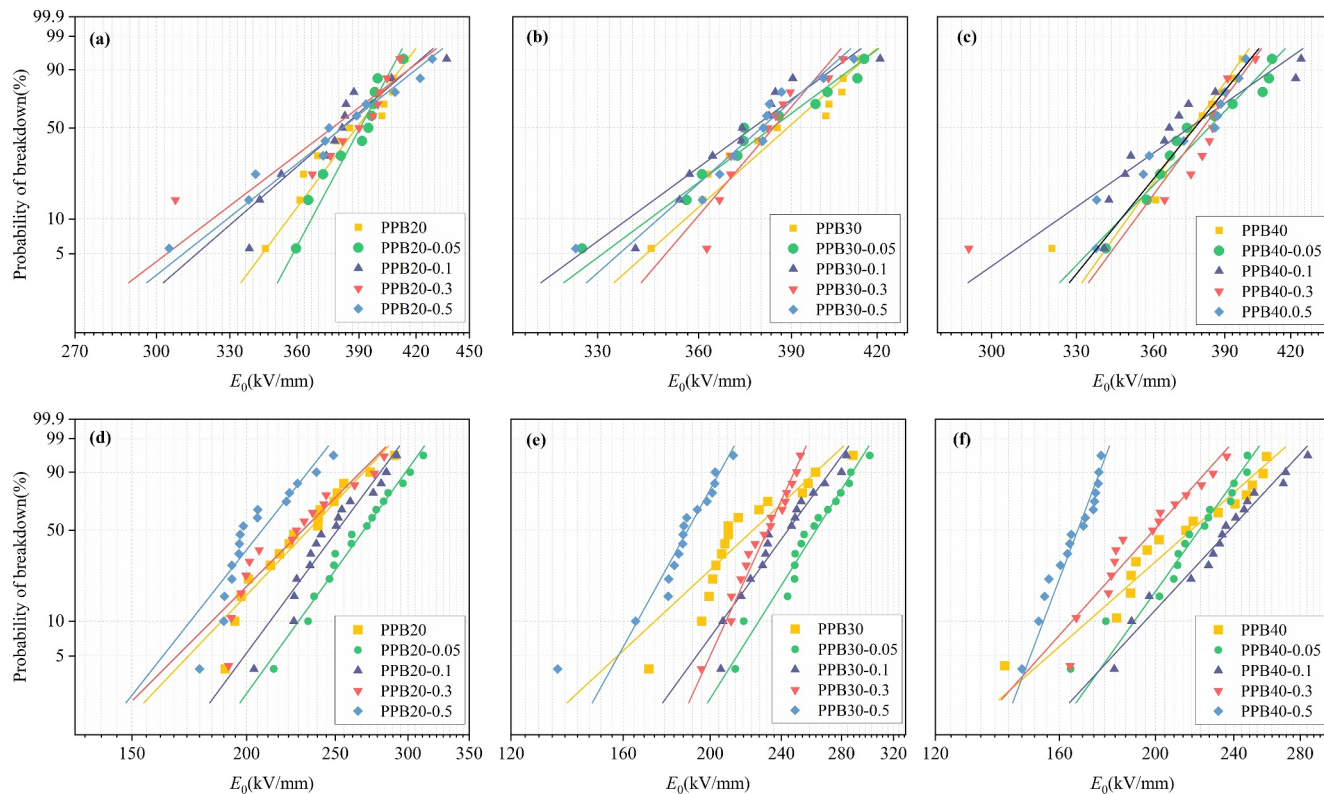


FIGURE 7 | Weibull distribution of DC breakdown strength for each sample at 25°C: (a) PPB20-y, (b) PPB30-y, (c) PPB40-y and at 90°C: (d) PPB20-y, (e) PPB30-y and (f) PPB40-y.

TABLE 6 | Parameters of the Weibull distribution of DC breakdown strength for each sample.

Samples	25°C		90°C		Rate of change of breakdown field strength at different temperatures (%)
	E_0 (kV/mm)	β_3	E_0 (kV/mm)	β_3	
PPB20	395.44	21.41	243.80	8.65	38.35
PPB20-0.05	395.32	30.00	277.38	11.48	29.83
PPB20-0.1	391.54	13.88	259.65	11.16	33.69
PPB20-0.3	388.22	12.16	240.77	8.25	37.98
PPB20-0.5	392.92	12.65	215.39	10.46	45.18
PPB30	388.99	22.30	234.40	7.49	40.15
PPB30-0.05	391.47	17.83	269.91	12.85	31.05
PPB30-0.1	385.16	17.52	252.17	11.13	34.52
PPB30-0.3	389.06	28.09	236.51	17.66	39.21
PPB30-0.5	386.64	21.25	193.55	14.62	49.79
PPB40	381.43	25.73	227.69	7.87	40.31
PPB40-0.05	390.40	19.09	228.06	12.48	41.58
PPB40-0.1	385.57	12.85	246.74	9.61	36.01
PPB40-0.3	386.07	24.90	206.78	9.91	46.44
PPB40-0.5	383.11	22.70	169.63	23.64	55.72

expansion of low-density regions [29]. The introduction of β -nucleator resulted in a slight improvement in the breakdown strength of each sample. A comparison of breakdown strengths among samples with different POE content under the same content of β -nucleator shows that the introduction of β -nucleator can inhibit the degradation of the DC breakdown strength associated with increasing elastomer content to some degree.

At 90°C, the variations in DC breakdown strength across the samples change more significantly. The increase in free volume within the amorphous region of PP at elevated temperatures facilitates charge migration and acceleration, resulting in a further decreasing breakdown strength. Notably, samples with β -nucleator content of less than 0.3 wt% demonstrate superior high-temperature resistance performance compared to a single toughened system. The DC breakdown strength of PPB30 decreased by about 40.15% with an increase in temperature, whereas PPB30-0.05 saw a reduction of 31.05%, and PPB30-0.1 reduced by 34.52%. However, when the β -nucleator content is excessive, the breakdown strength decreases significantly at 90°C. For instance, the breakdown strength of PPB30-0.5 decreased by 49.79%.

At 25°C, the shape parameters of all samples were relatively high, indicating that the breakdown data were more dispersed at this temperature. At 90°C, the shape parameters of the samples without the β -nucleator were the lowest. Upon the addition of the nucleating agent, as its concentration increased, the shape parameters of the samples generally exhibited an initial increase followed by a decrease, suggesting an improvement in the dispersion of the breakdown data. This indicates that with the nucleating agent, the differences in breakdown field strength across various breakdown sites are reduced, and the samples demonstrate greater structural stability at high temperatures. The results from the Weibull distribution parameter analysis confirm that the introduction of the β -nucleator enhances the structural

stability and intrinsic breakdown field strength of the samples at elevated temperatures.

The primary factors affecting short-term dielectric breakdown strength are the quantity of free electrons and the energy these electrons acquire in the electric field, whereas the interface between the crystalline and amorphous regions within each sample is crucial in determining the breakdown characteristics [30–32]. Combined with the previous analysis of structural characteristics and trap properties of each sample, it is clear that the introduction of the β -nucleator has improved the two-phase interface between POE and PP, and mitigated the effect of POE's inherent weaknesses on the breakdown strength of the blended system. Notably, optimising the crystallinity, the two-phase interface and conductivity–electric field stability of the system is most significant when the content of the β -nucleator is at 0.1 wt%, as shown in Figure 8. In that case, the DC breakdown strength of the sample is higher at 25°C and 90°C, the rate of change of DC breakdown strength is small and the temperature stability is high.

4 | Conclusion

This study explores the effect of adding β -nucleator to induce the formation of β -crystals in the iPP for elastomer-toughened insulation. The effect of varying concentrations of the β -nucleator and the elastomer on the aggregation state structure of PP insulation and its conductivity and breakdown strength characteristics in the DC electric field was investigated. The research results show the following:

1. Upon adding the nucleating agent, the crystallization characteristics of the samples have improved, and the trap energy level has been reduced. As the β -nucleator content gradually increases, the inhibition of the nucleation efficiency of the β -nucleator influenced by the elastomer is

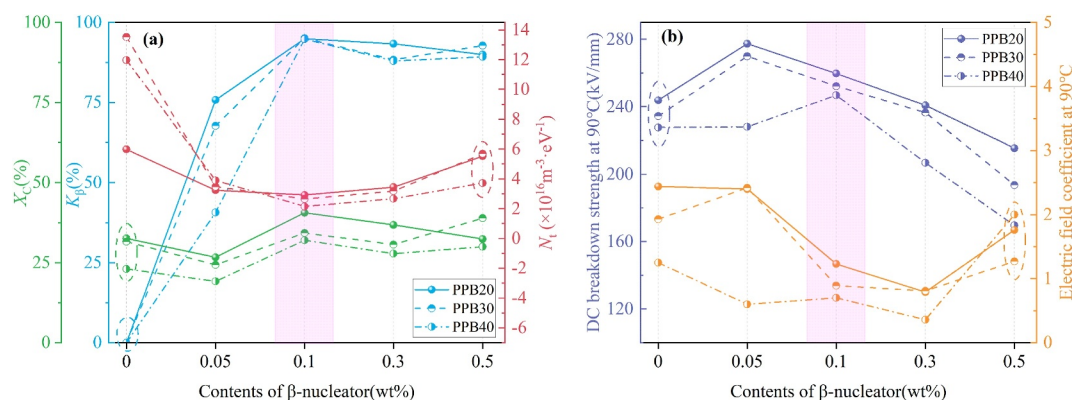


FIGURE 8 | Relationship between contents of β -nucleator and properties for each sample: (a) structural parameters, (b) DC characteristics.

weakened. Notably, at a β -nucleator content of 0.1 wt%, the PPB30-0.1 blend exhibits the highest crystallinity and β -crystalline proportion and the lowest trapped charge density. At this point, the two-phase interface structure of PP and POE demonstrates enhanced high-temperature stability.

- The addition of β -nucleator decreases the conductivity of the samples at 25°C. At the 0.1 wt% β -nucleator content, the electric field coefficient of PPB30-0.1 is reduced by 26.52% and 53.89% at 25°C and 90°C, respectively, compared with the single elastomer-toughened system, which indicates that the PPB30-0.1 blend exhibits low DC conductivity with high-electric field stability at both room and elevated temperatures. Furthermore, the DC breakdown strength of PPB30-0.1 is also high at 90°C, and the breakdown strength of the rate of decrease is lower by about 4.86% than that of the non-nucleator added system at high temperature.
- There are two notable effects after adding a β -nucleator for crystalline modification. Firstly, the unique structure characterised by the parallel arrangement of β -PP wafers and higher crystallinity strengthens the intermolecular interaction at the interface between the PP phase and the POE phase, and enhances the stability of the two-phase structure. This improvement effectively suppresses the effect of electric field changes on charge transport within the polymer. Secondly, the β -nucleator influences the degree of order of the elastomeric chain and its crystallinity in the composite phase, which will enhance the weak points and optimise the polymer structure of the phase and aggregation state. Thus, the DC electrical performances of elastomer-toughened PP insulation at room temperature are excellent, and the DC electrical properties at elevated temperatures show significant enhancement.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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